# Autonomous Exploration: Multi-Level SLAM Navigation

Charting the Unknown with Real-Time Precision & Sensor Fusion.

True autonomy requires more than just movement; it demands intelligent perception. Built upon the ROS (Robot Operating System) framework, this mobile robot integrates advanced SLAM (Simultaneous Localization and Mapping) algorithms to navigate and map complex, unknown environments.

## Sensor Fusion & Control:

Utilized a hybrid software architecture, employing C++ for low-level hardware interrupt handling and motor control, while Python managed high-level sensor data processing and trajectory planning.

## Advanced Mapping:

Deployed joint optimization algorithms to solve the "Loop Closure" problem, minimizing drift and ensuring high-fidelity map generation across multiple levels.

## User Interaction:

Developed a responsive web interface enabling real-time telemetry monitoring and remote navigation commands, bridging the gap between complex robotics backend and user accessibility.